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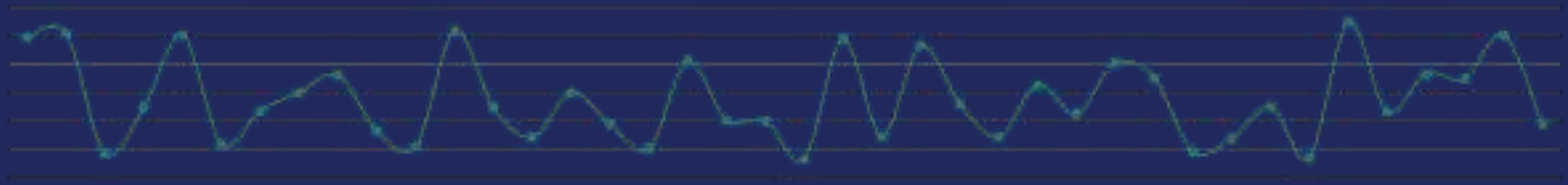
Time Series Analysis

Session 2 — Autoregressive Model (AR)



Prerequisites

What you should already be comfortable with before we start

Probability and Statistics
Distributions, expectation/variance, hypothesis testing

Linear Regression
OLS estimation, model assumptions, interpreting coefficients

Python Fundamentals
Comfortable writing and debugging basic scripts

pandas & notebooks
Loading, indexing, and plotting data in a Jupyter notebook

Course Content

A 6-week arc, from first principles to a final forecasting project

Week
1
Time Series Basics

Week
2
MA Models, Partial Autocorrelation, Notational Conventions

Week
3
Using ARIMA models to forecast future values

Week
4
Seasonal Models

Week
5
Smoothing and Decomposition Methods

Week
6
The Periodogram & project kickoff

Grading

How your final grade is composed

10%

Participation

Attendance & engagement in session

40%

Assignments

Hands-on tasks & presentations

50%

Final Project

Applied forecasting project on real data

Communication

Where course information lives and how to reach us

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Time Series Analysis Training Summer'26

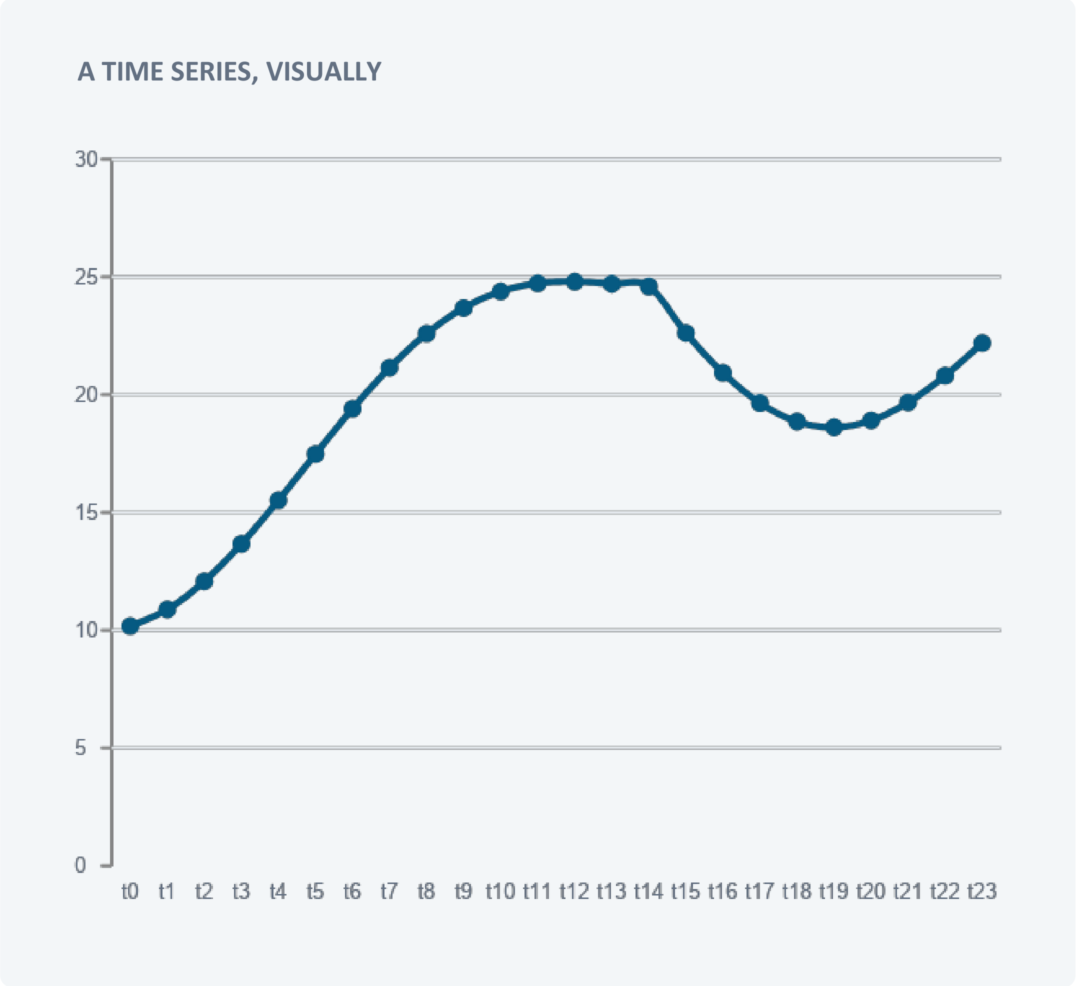
uirnsga

What Is a Time Series?

A univariate time series is a sequence of measurements of the same variable collected over time. Most often, the measurements are made at regular time intervals.

The key difference: order carries information. Unlike a typical regression dataset, you cannot shuffle the rows without losing something real.

- Daily stock closing prices
- Monthly unemployment rate
- Hourly temperature readings
- Quarterly GDP



Types of Models

Several families of models are used to describe and forecast time series

There are two basic types of “time domain” models.

- Models that relate the present value of a series to past values and past prediction errors - these are called ARIMA models (for Autoregressive Integrated Moving Average). We’ll spend substantial time on these.
- Ordinary regression models that use time indices as x-variables. These can be helpful for an initial description of the data and form the basis of several simple forecasting methods.

Cross-Sectional vs. Time Series Data

Cross-sectional

- Many units, observed at one point in time
- Observations assumed independent
- Row order carries no information
- Standard OLS assumptions typically apply

Time series

- One (or few) units, observed over time
- Observations are typically autocorrelated
- Row order carries real information
- Needs methods that model dependence over time

Important Characteristics to Consider First

Some important questions to first consider when first looking at a time series are:

T — **Trend**

Is there a trend, meaning that, on average, the measurements tend to increase (or decrease) over time?

S — **Seasonality**

Is there seasonality, meaning that there is a regularly repeating pattern of highs and lows related to calendar time such as seasons, quarters, months, days of the week, and so on?

O — **Outliers**

Are there outliers? In regression, outliers are far away from your line. With time series data, your outliers are far away from your other data.

V — **Constant Variance**

Is there constant variance over time, or is the variance non-constant?

C — **Long-run cycle**

Is there a long-run cycle or period unrelated to seasonality factors?

A — **Abrupt Changes**

Are there any abrupt changes to either the level of the series or the variance?

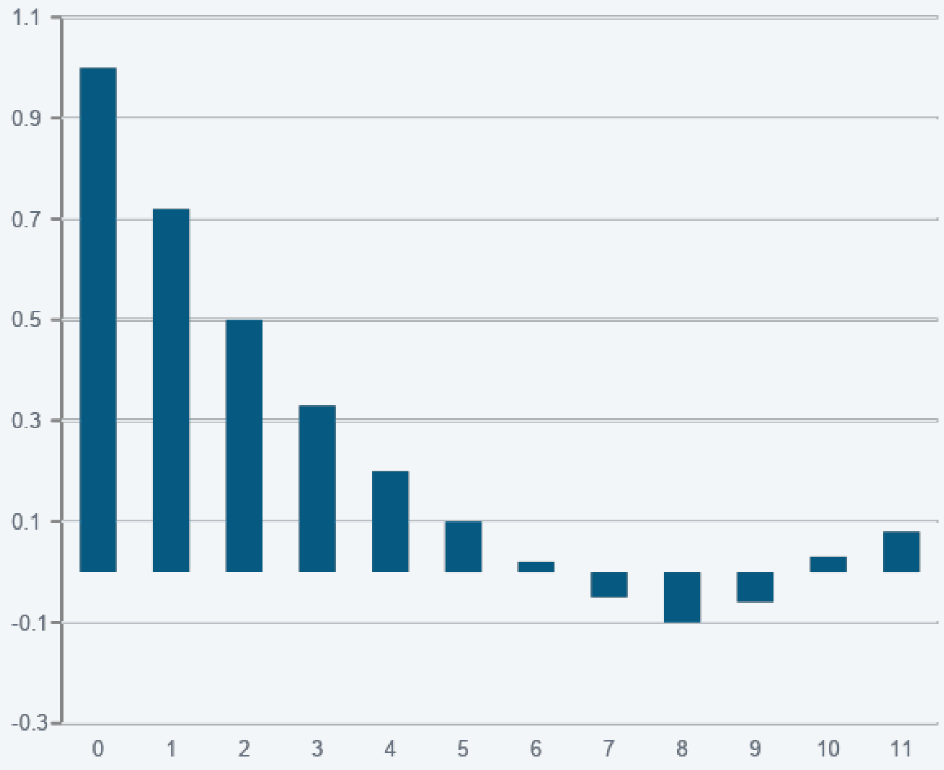
The Autocorrelation Function (ACF)

Measures how correlated a series is with its own past values, at each lag k.

$$\rho(k) = \frac{Cov(Y_t, Y_{t-k})}{Var(Y_t)}$$

Used to: spot repeating patterns, help choose model order, and check whether a series behaves like white noise.

A CORRELOGRAM (ACF BY LAG)



White Noise

A sequence of uncorrelated random variables and cannot be predicted.

μ **Zero mean**

σ^2 **Constant variance**

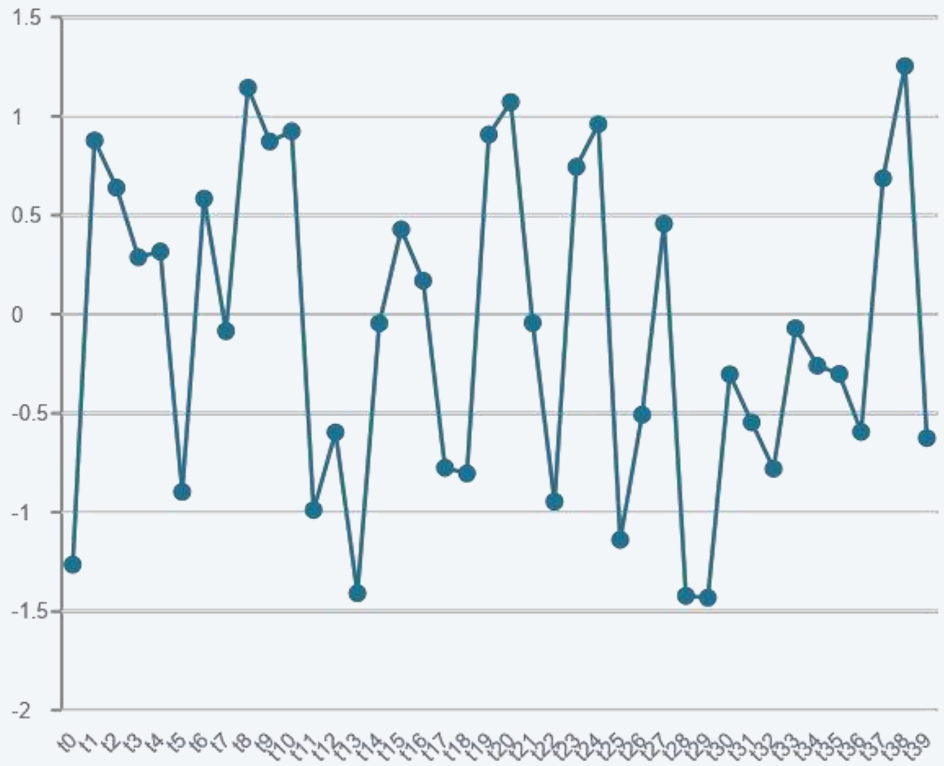
Same spread at every point in time

ρ **Zero autocorrelation**

No relationship between values at any lag

Why it matters: *the residuals of a good model should look like white noise.*

WHAT WHITE NOISE LOOKS LIKE



Stationarity

Stationarity is a property of a Time Series

- The time series which has constant mean, variance and autocorrelation is called stationary time series. It is recommended to have the stationary time series for better analysis.
- The predictions on non-stationary series may give wrong values.
- It is possible to make nearly any time series stationary by applying a suitable transformation. Most statistical forecasting methods are designed to work on a stationary time series.
- The first step in the forecasting process is typically to do some transformation to convert a non-stationary series to stationary.

Stationarity

Weak (covariance) stationarity requires three things to hold over time

μ

Constant mean
The average level doesn't drift up or down over time

σ^2

Constant variance
The spread of values doesn't grow or shrink over time

$\gamma(k)$

Lag-dependent covariance
Covariance between Y_t and Y_{t-k} depends only on k, not on t

Why it matters: most classical time series models assume stationarity. Non-stationary series are typically transformed before modeling.

How to test for Stationarity?

- Split the series into 2 or more contiguous parts and computing the summary statistics like the mean, variance and the autocorrelation. If the stats are quite different, then the series is not likely to be stationary.
- Quantitatively method to determine if a given series is stationary or not, is using statistical tests called ‘Unit Root Tests’. There are multiple variations of this, where the tests check if a time series is non-stationary and possess a unit root.

There are multiple implementations of Unit Root tests like:

- Augmented Dickey Fuller test (ADF Test)
- Kwiatkowski-Phillips-Schmidt-Shin (KPSS Test)
- Philips Perron test (PP Test)

How to make a Time Series Stationary?

Difference Transform

It is subtracting previous value with current value. It is done to remove the dependency of values on time.

Log Transform

Take the log of the series, this can help stabilize the variance of the series which is desirable for stationary analysis.

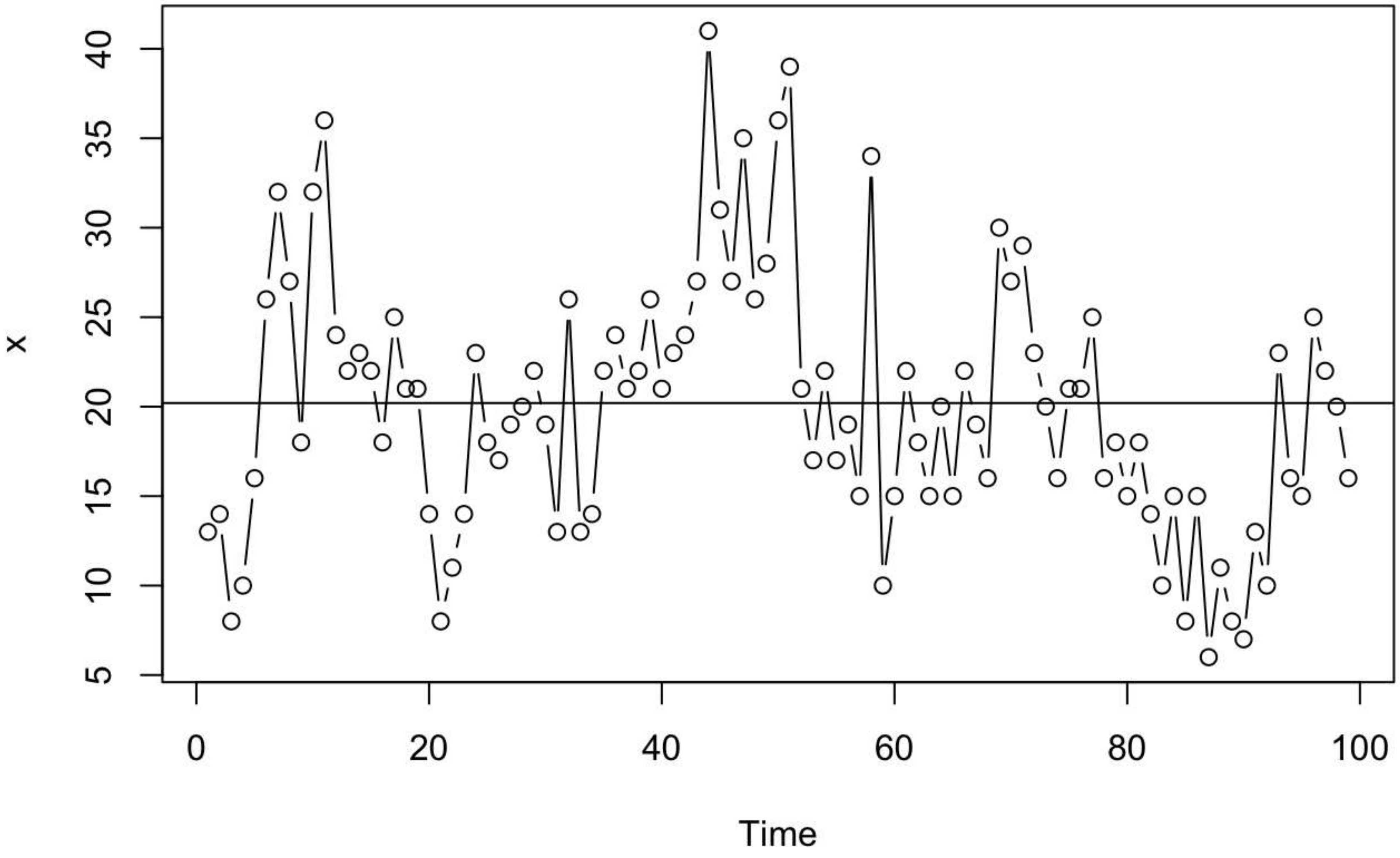
Nth Root Transform

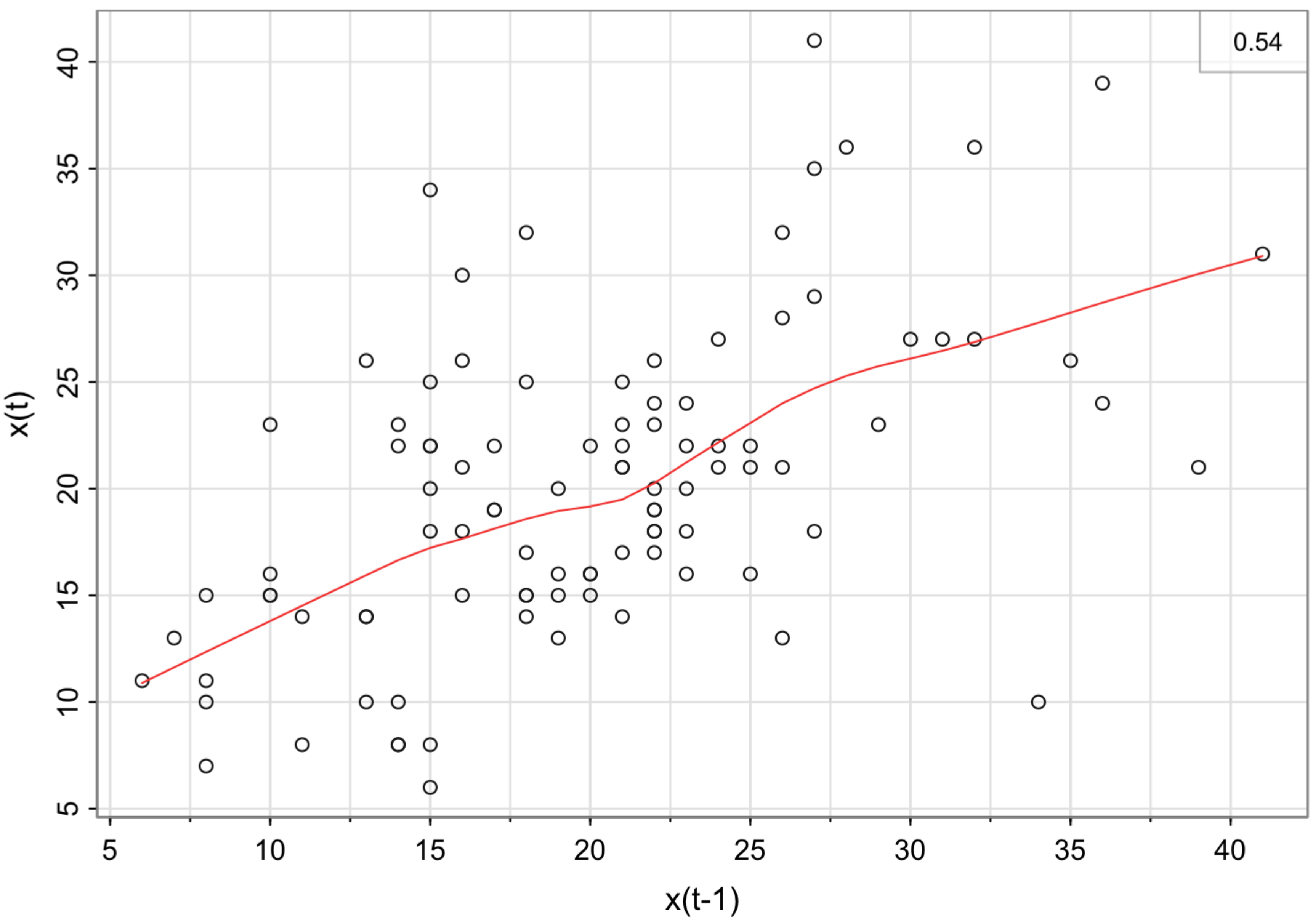
Take the nth root of the series this can help stabilize the variance by reducing the magnitude of extreme values.

De-trending & De-seasonality

Removing trend and seasonality

The following plot is a **time series plot** of the annual number of earthquakes in the world with seismic magnitude over 7.0, for 99 consecutive years.




0.54
x(t)
40
35
30
25
20
15
10
5
x(t-1)
5
10
15
20
25
30
35
40

Setting Up an AR(1)

Annual counts of major earthquakes (magnitude 7+), 99 consecutive years

1. **No consistent trend**
   The series wanders above and below its long-run mean rather than trending steadily up or down.

2. **No seasonality**
   The data are annual, so there is no calendar-based repeating pattern to account for.

3. **Positive lag-1 relationship**
   Plotting x_t against x_{t-1} (its lag 1 value) shows a moderate positive linear association.

Why it matters: A positive lag-1 relationship is exactly what an AR(1) model needs — it predicts the present value using only the value one time step before it.

The AR(1) Model

Autoregressive model of order 1 — predicts the present using the immediately preceding value

$$x_t = \delta + \varphi_1 x_{t-1} + w_t$$

1. **White noise errors**
   w_t is independent and identically distributed with mean 0 and constant variance σ^2_u.

2. **Errors independent of x**
   w_t is independent of x_{t-1} and all earlier values of the series.

3. **Stationarity needs $|\phi_1| < 1$**
   This keeps the variance of x_t finite and constant — without it the process is not stationary.

Why it matters: AR(1) looks like an ordinary regression, but the “predictor” x_{t-1} is itself random — it's a past value of the same series, not a fixed x-variable.

Properties of the AR(1) Model

Mean, variance, and autocorrelation follow directly from the model equation

μ

Mean: $E(x_t) = \dfrac{\delta}{(1 - \varphi_1)}$

σ²

Variance: $Var(x_t) = \dfrac{\sigma_w^2}{(1 - \varphi_1{}^2)}$

ρ

Autocorrelation at lag h: $\rho_h = \varphi_1{}^h$

ϕ_1 is both the AR(1) coefficient and the lag-1 autocorrelation

Why it matters: These formulas only hold when $|\phi_1| < 1$. The ACF of a true AR(1) process decays geometrically — that geometric-decay shape is the pattern we look for in a sample ACF.

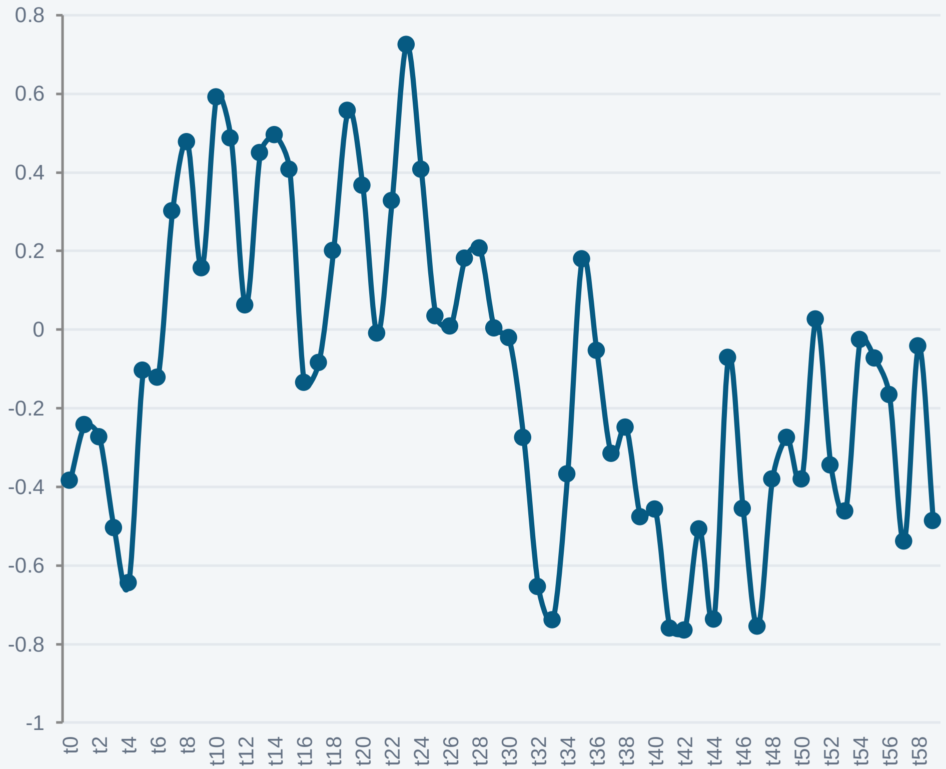
AR(1) — Autoregressive Model of Order 1

$$X_t = \delta + \varphi X_{t-1} + W_t$$

ϕ controls how much the series “remembers” its own past:

$\lvert\varphi\rvert < 1$	Stationary — shocks fade out over time
$\varphi = 1$	Random walk — non-stationary
$\lvert\varphi\rvert > 1$	Explosive — shocks grow without bound

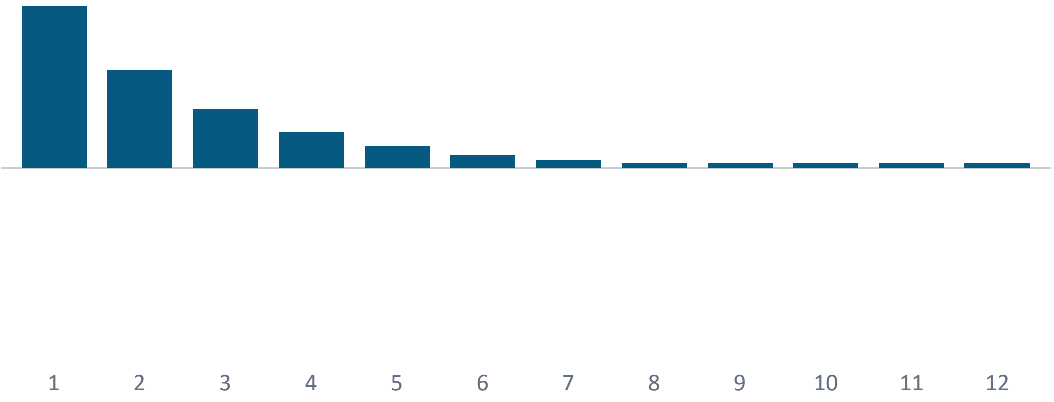
A STATIONARY AR(1) PATH (ϕ ≈ 0.7)



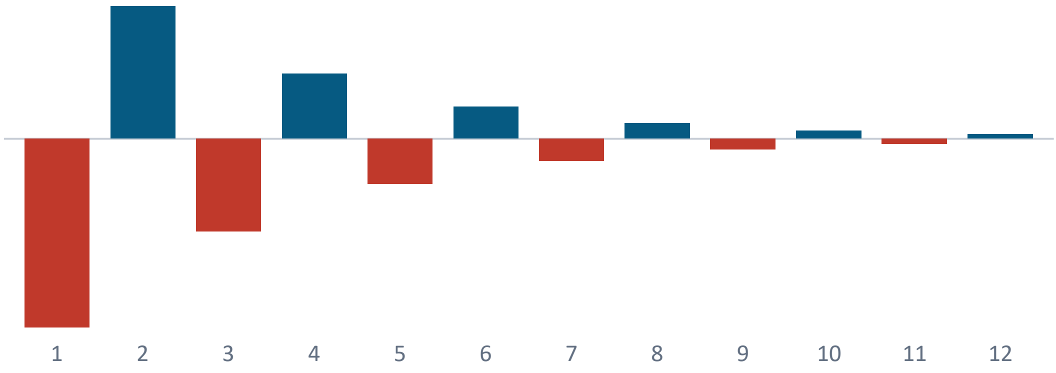
Pattern of the ACF for AR(1)

The sign of φ_1 determines how the correlogram decays

$\phi_1 = 0.6$ (positive)



$\phi_1 = -0.7$ (negative)



Why it matters: Positive ϕ_1 gives an ACF that tapers smoothly to 0, all on one side. Negative ϕ_1 gives an ACF that also tapers to 0, but alternates sign at every lag.

The Partial Autocorrelation Function (PACF)

How to select the right order of AR to forecast a value in time series?

How many lagged values will I include in my regression equation to help me predict the next in time series?
That’s when **PACF** plot steps in!

It measures the correlation between a series and its lag k, after removing the effect of the shorter lags in between.

$$\varphi(k) = \mathrm{corr}(Y_t, Y_{t-k} \mid Y_{t-1}, \ldots, Y_{t-k+1})$$

Used to: isolate the direct effect of each lag, and help decide the order of an AR model.

A PACF PLOT (PARTIAL CORRELATION BY LAG)